

POLITECNICO
MILANO 1863

Introduction about sensors

SENSOR SYSTEMS

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TRANSDUCER = **device which transforms energy from one type to another,**
even if both energy types are in the same domain.

Typical energy domains are mechanical, electrical, chemical, magnetic, optical and thermal.

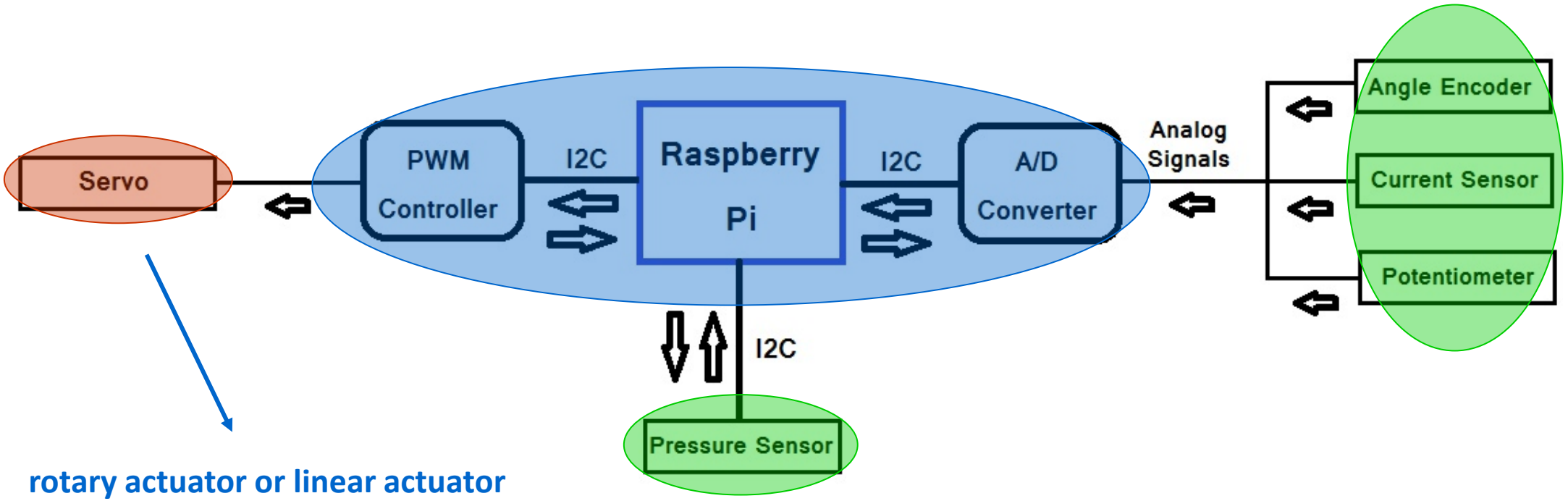
Transducer can be further divided into **Sensors** and **Actuators**

SENSOR = **device which monitor a parameter of a system,**
hopefully without disturbing that parameter.

The specific input could be light, heat, motion, moisture, pressure,
or any one of a great number of other environmental phenomena.
The output is generally a signal that is converted to human-readable
display at the sensor location or transmitted electronically over a network
for reading or further processing.

ACTUATOR = **component of machines which is responsible for moving**
or controlling a mechanism or system.

actuator ⟸ electronics ⟸ sensors



rotary actuator or linear actuator
that allows for precise control
of angular or linear position,
velocity and acceleration

Classification based on **physical phenomena**

- Optical: visible/IR light (photodiode, CCD, CMOS APS, infrared sensor)
- Thermal: temperature (RTD, thermistor, thermocouple…)
- Magnetic: magnetic field (Hall effect sensor, magneto-resistive sensor)
- Mechanical: strain (strain gauge), force (piezo-electric sensor), displacement and distance (capacitive, inductive, acoustic, optical), acceleration and orientation (MEMS), viscosity, pressure, etc.
- Chemical: pH (pH-meter)

Classification based on **measuring mechanism**

- Resistive sensing
- Capacitive sensing
- Inductive sensing
- Piezoelectricity
- Hall Effect
- MEMS

Which sensors will we study?

Physical phenomena	Sensor
Light and images	- Photodiode - CCD - CMOS sensor
Temperature	- Resistance Temperature Detectors (RTD) - Thermistors (NTC and PTC) - Thermocouple - Infrared thermometer
Magnetic Field	- Hall effect sensors - Magneto-resistances
Strain, force, pressure	- Strain gauges - Piezoelectric sensors
Displacement, proximity, distance and rotation	- Capacitive sensors - Inductive sensors - Acoustic sensors - Optical sensors
Acceleration and Orientation	- Accelerometers (MEMS) - Gyroscopes (MEMS)
Sound	- Microphone

LAB 8: Accelerometer

Sensitivity: the ratio between the change in the output signal to a small change in input physical signal.
Slope of the input-output fit line.

$$S = \frac{\partial\ out}{\partial\ in}$$

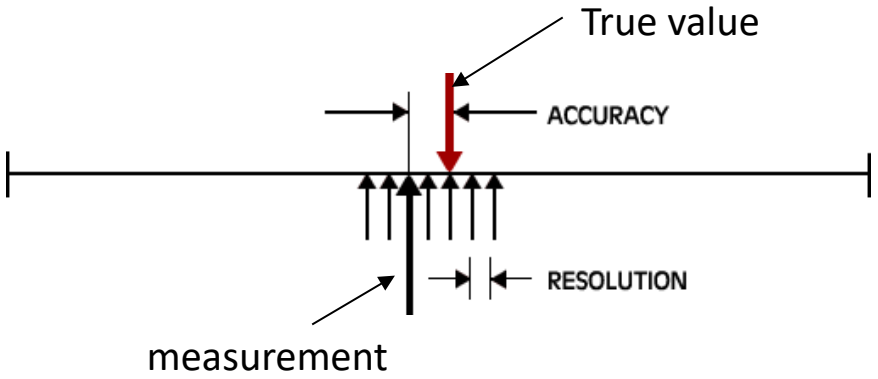
Resolution (LSB): the smallest increment of measure that a device can make (Least Significant Bit).

Full Scale Range (FSR): the maximum interval of measure that a device can cover.

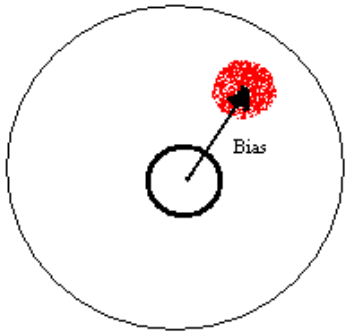
Number of bits (n): in sensors with digital output 2^n is the number of levels in which the FSR is divided.

$$\boldsymbol{2^n = \frac{FSR}{LSB}}$$

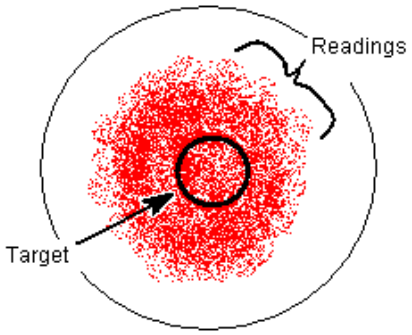
Accuracy: error between the result of a measurement and the true value being measured.



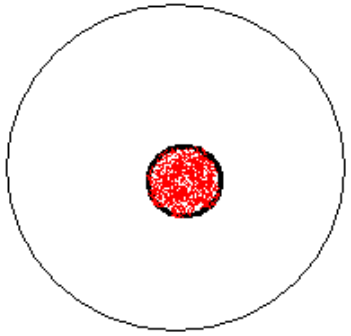
Repeatability/Precision: the ability of the sensor to output the same value for the same input over a number of trials.



Precision without accuracy



Accuracy without precision

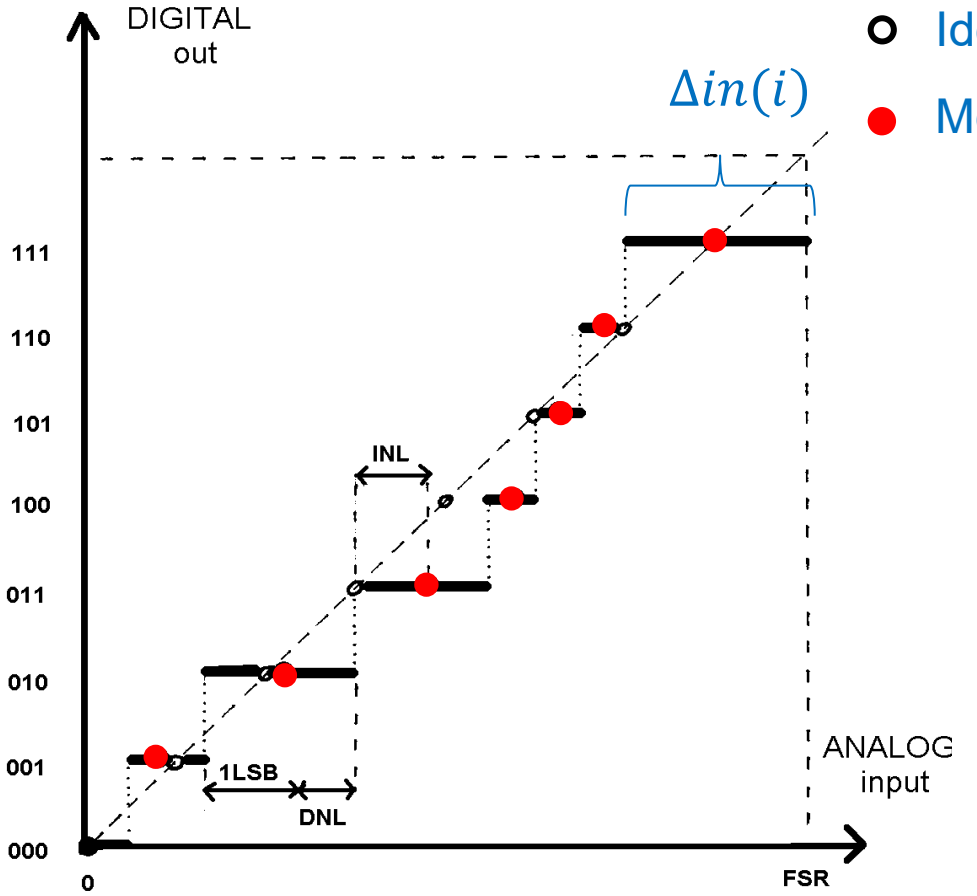


Precision and accuracy

Sensors' characteristics (3)

Linearity: the deviation of the output from a best-fit straight line for a given range of the sensor.

Differential Non-Linearity (DNL) and Integral Non-Linearity (INL)



Ideal value

Measured value

$$DNL(i) = \Delta in(i) - LSB$$

$$INL(i) = \sum_{k=1}^{i} DNL(k)$$

DNL: Differential Non-Linearity

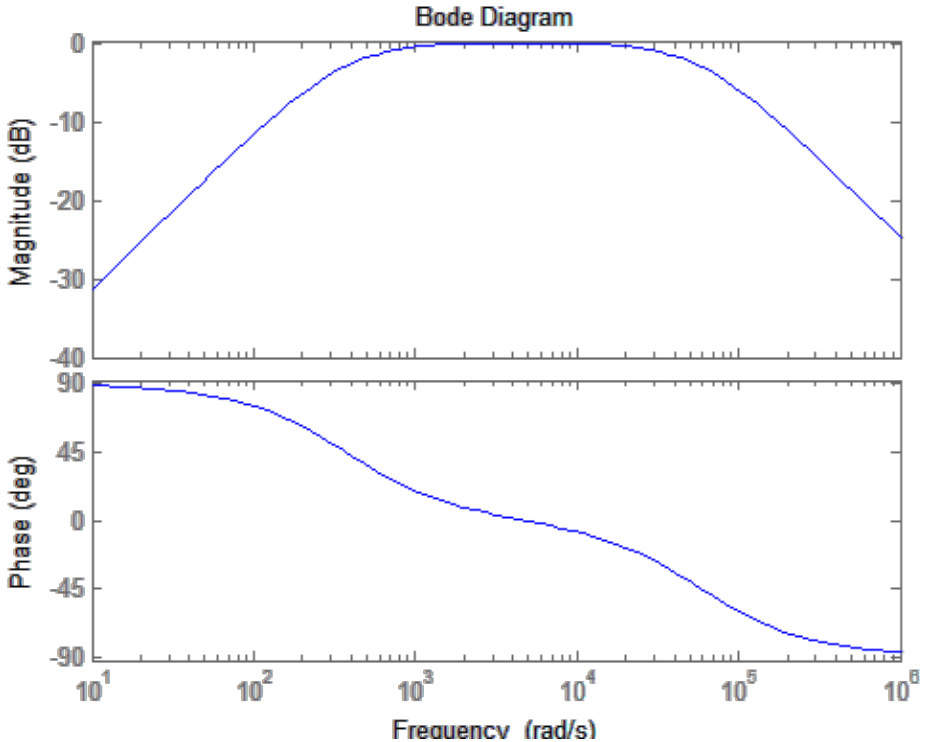
difference of the real step, compared to the ideal one (i.e. 1 LSB)

INL: Integral Non-Linearity

distance between real and ideal output voltage

Transfer Function (Frequency Response): The relationship between physical input signal and electrical output signal, which may constitute a complete description of the sensor characteristics as a function of frequency.

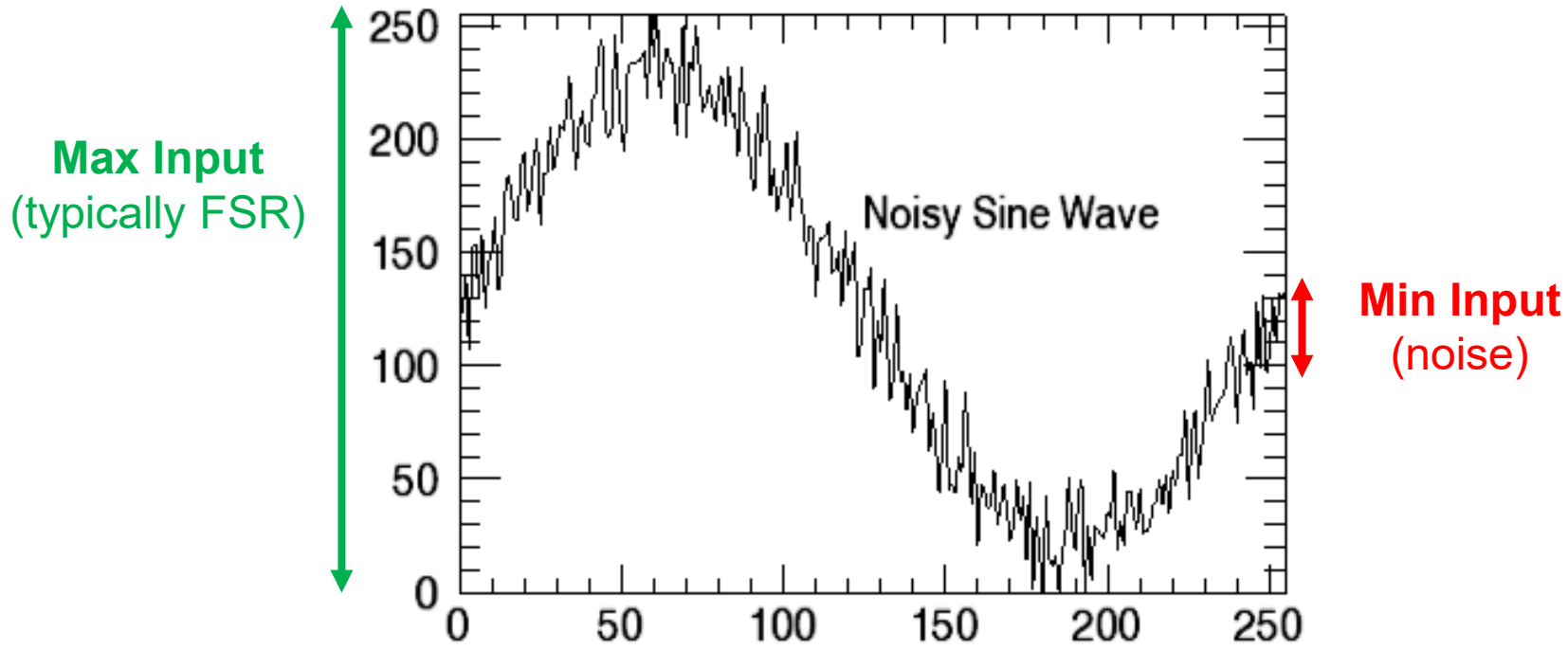
Bandwidth: the frequency range between the lower and upper cutoff frequencies, within which the sensor transfer function is constant gain or linear.



Noise: random fluctuation in the measured value. It is quantified with its rms (root mean square) value.

Dynamic Range: the ratio of maximum recordable input amplitude to minimum input amplitude:

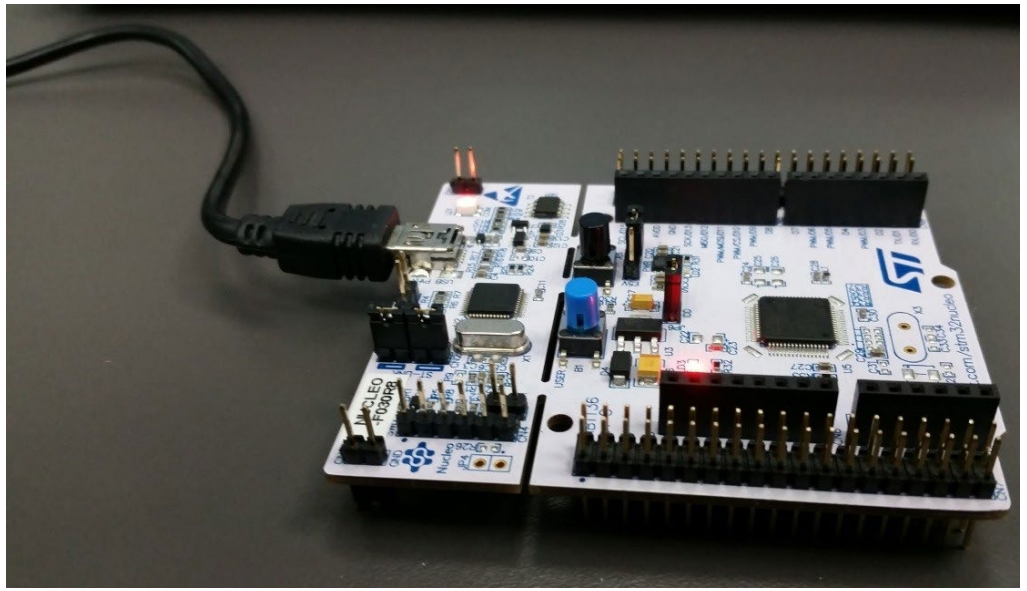
$$DR = 20 \cdot Log\left(\frac{Max\ Input\ Amplitute}{Min\ Input\ Amplitude}\right)$$ (expressed in dB)



Analog sensor readout

- ADC + microcontroller
  many microcontrollers have a built-in ADC
  (typically, 8-bit to 12-bit or even 16-bit)

LAB 5 and LAB 6: ADC



- Data Acquisition Cards (DAQ)
  - ✓ PC card or external devices with analog and digital I/O
  - ✓ interface through LabVIEW or user-generated code



Smart sensor

= sensor with built-in signal processing & communication
(e.g., combining a “dumb sensor” and ADC / processing)

Digital output:

- parallel bus

- serial I/O

  synchronous (with clock): SPI, I^2C
  (must match byte format, stop/start bits, parity check, etc.)

  - Serial Peripheral Interface (SPI)

    1 clock + 1 bidirectional data + 1 chip select/enable

  - Inter Integrated Circuit (I^2C)

    1 clock + 1 data

  asynchronous (no clock): one wire
  (must match baud rate and bit width, transmission protocol, etc.)

  - Universal Asynchronous Receiver Transmitter (UART)

Sensors can exhibit non-ideal effects

- offset: nominal output ≠ nominal parameter value, with a fixed difference
- nonlinearity: output not linear with parameter changes
- cross parameter sensitivity: secondary output variation with other parameters (e.g., temperature drifts)

Calibration = adjusting output to match parameter

- analog signal conditioning
- look-up table
- digital calibration

  $T = a + bV + cV^2$,

  - T= temperature; V=sensor voltage;
  - a,b,c = calibration coefficients